

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Video Projector
Manufacturer: BenQ
Firmware Version: N/A
Model(s): MW612

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
2.08.0002-b002	1.4.2

Version History

Module Version	Date	Notes
1_0_0_0	6/25/2018	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- Device only responses to one query at a time, send update commands 3 seconds apart

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='MW612')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='MW612')</code>

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command 3DFormat	Value 'Off' 'Side by Side'	Value 'Top Bottom'	Value 'Frame Sequential'
# 3DFormat example InterfaceName.Set('3DFormat', 'Off')			
Command 3DInvert	Value 'On'	Value 'Off'	
# 3DInvert example InterfaceName.Set('3DInvert', 'On')			
Command AspectRatio	Value '4:3' 'Auto'	Value '16:9' 'Real'	Value '16:10'
# AspectRatio example InterfaceName.Set('AspectRatio', '4:3')			
Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute example InterfaceName.Set('AudioMute', 'On')			
Command AutoImage ¹			
# AutoImage example InterfaceName.Set('AutoImage', 'On')			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command Input	Value 'Computer/YPbPr' 'Video'	Value 'HDMI 1' 'S-Video'	Value 'HDMI 2/MHL'
# Input example InterfaceName.Set('Input', 'Computer/YPbPr')			
Command LampMode	Value 'Normal'	Value 'Eco'	Value 'SmartEco'
# LampMode example			

Global Scripter Module Communication Sheet

InterfaceName.Set('LampMode', 'Normal')			
Command MenuNavigation	Value 'Menu On' 'Down' 'Enter'	Value 'Menu Off' 'Right'	Value 'Up' 'Left'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Menu On')			
Command PictureMode	Value 'Presentation' 'DICOM' 'User 2'	Value 'sRGB' 'Vivid'	Value 'Bright' 'User 1'
# PictureMode example InterfaceName.Set('PictureMode', 'Presentation')			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command VideoMute	Value 'On'	Value 'Off'	
# VideoMute example InterfaceName.Set('VideoMute', 'On')			
Command Volume	Value 'Up'	Value 'Down'	
# Volume example InterfaceName.Set('Volume', 'Up')			

¹ Only available for Computer/YPbPr

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AspectRatio	Value '4:3'	Value '16:9'	Value '16:10'
	'Auto'	'Real'	
# AspectRatio examples InterfaceName.Update('AspectRatio') Value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute examples InterfaceName.Update('AudioMute') Value = InterfaceName.ReadStatus('AudioMute') InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
# ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze examples InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)			
Command Input	Value 'Computer/YPbPr'	Value 'HDMI 1'	Value 'HDMI 2/MHL'

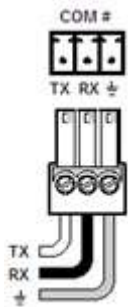
Global Scripter Module Communication Sheet

	'Video'	'S-Video'	
# Input examples InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command LampMode	Value 'Normal'	Value 'Eco'	Value 'SmartEco'
# LampMode examples InterfaceName.Update('LampMode') Value = InterfaceName.ReadStatus('LampMode') InterfaceName.SubscribeStatus('LampMode', None, FeedbackHandler)			
Command LampUsage	Value —		
# LampUsage examples InterfaceName.Update('LampUsage') Value = InterfaceName.ReadStatus('LampUsage') InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)			
Command PictureMode	Value 'Presentation'	Value 'sRGB'	Value 'Bright'
	'DICOM'	'Vivid'	'User 1'
	'User 2'	'3D'	
# PictureMode examples InterfaceName.Update('PictureMode') Value = InterfaceName.ReadStatus('PictureMode') InterfaceName.SubscribeStatus('PictureMode', None, FeedbackHandler)			
Command Power	Value 'On'	Value 'Off'	
# Power examples InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)			
Command VideoMute	Value 'On'	Value 'Off'	
# VideoMute examples InterfaceName.Update('VideoMute') Value = InterfaceName.ReadStatus('VideoMute') InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)			
Command VolumeStatus	Value 0 – 10		
# VolumeStatus examples InterfaceName.Update('VolumeStatus') Value = InterfaceName.ReadStatus('VolumeStatus') InterfaceName.SubscribeStatus('VolumeStatus', None, FeedbackHandler)			

Cable and Adapter Requirements

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device

Serial communication**Port Type:** RS-232**Baud Rate:** 115200**Data Bits:** 8**Parity:** None**Stop Bits:** One**Flow Control:** None**Pin Assignments Diagram**

Signal	Main Cable	Pin	Signal
TxD		2	TxD
RxD		3	RxD
GND		5	GND



Appendix A. Set Commands

3D Format Frame Sequential	\x0D*3d=fs#\x0D
3D Format Off	\x0D*3d=off#\x0D
3D Format Side by Side	\x0D*3d=sbs#\x0D
3D Format Top Bottom	\x0D*3d=tb#\x0D
3D Invert Off	\x0D*3d=da#\x0D
3D Invert On	\x0D*3d=iv#\x0D
Aspect Ratio 16:10	\x0D*asp=16:10#\x0D
Aspect Ratio 16:9	\x0D*asp=16:9#\x0D
Aspect Ratio 4:3	\x0D*asp=4:3#\x0D
Aspect Ratio Auto	\x0D*asp=AUTO#\x0D
Aspect Ratio Real	\x0D*asp=REAL#\x0D
Audio Mute Off	\x0D*mute=off#\x0D
Audio Mute On	\x0D*mute=on#\x0D
Auto Image None	\x0D*auto#\x0D
Freeze Off	\x0D*freeze=off#\x0D
Freeze On	\x0D*freeze=on#\x0D
Input Computer/YPbPr	\x0D*sour=RGB#\x0D
Input HDMI 1	\x0D*sour=hdm1#\x0D
Input HDMI 2/MHL	\x0D*sour=hdm2#\x0D
Input S-Video	\x0D*sour=svid#\x0D
Input Video	\x0D*sour=vid#\x0D
Lamp Mode Eco	\x0D*lampm=eco#\x0D
Lamp Mode Normal	\x0D*lampm=lnor#\x0D
Lamp Mode SmartEco	\x0D*lampm=seco#\x0D
Menu Navigation Down	\x0D*down#\x0D
Menu Navigation Enter	\x0D*enter#\x0D
Menu Navigation Left	\x0D*left#\x0D
Menu Navigation Menu Off	\x0D*menu=off#\x0D
Menu Navigation Menu On	\x0D*menu=on#\x0D
Menu Navigation Right	\x0D*right#\x0D
Menu Navigation Up	\x0D*up#\x0D
Picture Mode Bright	\x0D*appmod=bright#\x0D
Picture Mode DICOM	\x0D*appmod=dicom#\x0D
Picture Mode Presentation	\x0D*appmod=preset#\x0D
Picture Mode User 1	\x0D*appmod=user1#\x0D
Picture Mode User 2	\x0D*appmod=user2#\x0D

**Global Scripter Module
Communication Sheet**

Picture Mode Vivid	\x0D*appmod=vivid#\x0D
Picture Mode sRGB	\x0D*appmod=srgb#\x0D
Power Off	\x0D*pow=off#\x0D
Power On	\x0D*pow=on#\x0D
Video Mute Off	\x0D*blank=off#\x0D
Video Mute On	\x0D*blank=on#\x0D
Volume Down	\x0D*vol=-#\x0D
Volume Up	\x0D*vol=+#\x0D

Appendix B. Update Commands

Power	\x0D*pow=?#\x0D
Aspect Ratio	\x0D*asp=?#\x0D
Audio Mute	\x0D*mute=?#\x0D
Freeze	\x0D*freeze=?#\x0D
Input	\x0D*sour=?#\x0D
Lamp Mode	\x0D*lampm=?#\x0D
Lamp Usage	\x0D*ltim=?#\x0D
Picture Mode	\x0D*appmod=?#\x0D
Video Mute	\x0D*blank=?#\x0D
Volume Status	\x0D*vol=?#\x0D